and chemical role. Indeed, it's hard for us water-besotted creatures to imagine life without water. If on our planet the origin of life took less than a hundred million years, is there any chance that on Titan it took a thousand? With tholins mixed into liquid water—even for only a thousand years—the surface of Titan may be much further along toward the origin of life than we thought.

DESPITE ALL THIS, we understand pitifully little about Titan. This was brought home forcefully to me at a scientific symposium on Titan held in Toulouse, France, and sponsored by the European Space Agency (ESA). While *oceans* of liquid water are impossible on Titan, oceans of liquid hydrocarbons are not. Clouds of methane (CH_4), the most abundant hydrocarbon, are expected not far above the surface. Ethane (C_2H_6), the next most abundant hydrocarbon, must condense out at the surface in the same way that water vapor becomes a liquid near the surface of the Earth, where the temperature is generally between the freezing and melting points. Vast oceans of liquid hydrocarbons should have accumulated over the lifetime of Titan. They would lie far beneath the haze and clouds. But that doesn't mean they would be wholly inaccessible to us—because radio waves readily penetrate the atmosphere of Titan and its suspended, slowly falling fine particles.

In Toulouse, Duane O. Muhleman of the California Institute of Technology described to us the very difficult technical feat of transmitting a set of radio pulses from a radio telescope in California's Mojave Desert, so they reach Titan, penetrate through the haze and clouds to its surface, are reflected back into space, and then returned to Earth. Here, the greatly enfeebled signal is picked up by an array of radio telescopes near Socorro, New Mexico. Great. If Titan has a rocky or icy surface, a radar pulse reflected off its surface should be detectable on Earth. But if Titan were covered with hydrocarbon oceans, Muhleman shouldn't see a thing: Liquid hydrocarbons are black to these radio waves, and no echo would have been returned to Earth. In fact, Muhleman's giant radar system